A Comparative Study of Designer and User Perceptions on the Usability, Design Quality, and Challenges of AI-Assisted vs. Manually Designed UI/UX

Muhammad Iqbal bin Zakaria
Asia Pacific University
Iqbalzakaria055@gmail.com

Abstract

The rapid growth of artificial intelligence has resulted in everything AI-generated including the process of designing the user interface and user experience. The integration of large language models and image models has greatly contributed to how digital products are conceptualised and built. While the whole world is accepting this technological advancement, there are several critical concerns that are not being paid attention to; Generative AI systems may produce unlimited synthetic media and disinformation, severely undermining information integrity in design outputs. Moreover, the normalization of reliance on AI tools risks eroding designer's foundational skills that might cause skill gaps among emerging practitioners. Furthermore, integrating AI into established design workflows may require costly overhauls of existing processes, software, and team structures. This might disrupt the productivity during the transition. Crucially, how users actually perceive and prefer AI-generated interfaces over manually designed ones amid these concern remains poorly understood. This study investigates user preferences regarding the usability and experiential quality of AI-generated versus manually designed UI/UX interfaces, and how awareness of above statements influences the preferences like whether or not AI need interference with human touch or solely on its own. A comparative user study was conducted with [N] participants with different backgrounds such as normal users and designers who evaluated paired AI-generated and manually designed interface prototypes across usability, aesthetic appeal, and trust dimensions using standardized questionnaires.

Keywords

Generative AI, UI/UX Design, User Perception, Usability Testing, Human-AI Co-Creation, Information Integrity, Design Automation.

1. Introduction

This evolutionary era of Artificial Intelligence (AI) has significantly transformed the landscape of digital product development, particularly within the domain of User Interface (UI) and User Experience (UX) design. Generative AI is software algorithms, specifically Large Language Models (LLMs) (e.g. ChatGPT, Gemini, and Claude) and advanced Image Generation Models (e.g. Midjourney and DALL-E) that dynamically synthesize text, layouts, code, and visual assets based on user-prompted instruction. Nowadays industries are taking bigger steps to integrate these automated tools into corporate and freelance design workflows. Generative AI has significantly accelerated product production cycles and lowered the barrier entry for digital interface creation by enabling rapid prototyping, instantaneous wireframing, and automatic aesthetic generation.

This massive shift, often seems as harmless, are fundamentally disrupting the traditional design methodologies that historically relied on human touch, iterative testing, and manual craft. While the giant tech companies are competing to adopt this technological advancement, a wave of critical concerns has emerged. Many generative AI engines offer unlimited and unrestricted proliferation which carries the inherent risk of producing vast quantities of synthetic media, which threatens to introduce algorithmic bias and disinformation into design outputs, thereby compromising information integrity. Furthermore, over-reliance on automated tools risk eroding the foundational critical-thinking and technical execution skills of emerging UI/UX practitioners. This will create a dangerous skill gap in the workforce. Operationally, integrating AI into established corporate ecosystem requires costly overhauls of legacy software, workflows, and team structures. This creates friction and temporary drops in productivity.

Crucially, in this era of technical, educational, and operational updates, the most important component of digital ecosystem remains overlooked, which is the end-user. Failure and success of any digital application depend on user adoption, perception, and experiential trust, rather than the speed of its production pipeline. The data relating to how users actually perceive, interact, and trust these AI-assisted digital products remains scarce even though the benchmark of AI advancements are growing. This critical gap in existing knowledge presents a major risk for product teams. The absence of clear understanding of the end user preferences regarding the usability and aesthetic quality of AI-designed versus human-designed layouts caused the industry to have risks when deploying highly optimized, automated software that fails to resonate with human experience.

2. Problem Statement

The preferred and ideal digital product development pipeline, technological advancements should optimize production efficiency without ignoring the interface quality, workforce competency, or user trust. However, the rapid growth and wide spreading use of generative AI, specifically in UI/UX workflows has introduced a significant systemic risk.

Information integrity rules are being broken when these automated systems generate synthetic media and potential disinformation (Kim et al., 2024). Individuals who aim to spread disinformation to manipulate public opinion, control information space, and influence political processes, are opposing great dangers to certain organisations e.g. political and public discourse. AI could potentially have a destructive influence when fake news websites or voiced AI-generated virtual personalities are used to spread manipulative information. While AI are able to provide a very much comprehensible and accurate information compared to most people, the dangers it holds to generate more persuasive disinformation cannot be tolerated (Marushchak et al., 2024). Moreover, skill gaps are increasing between the UI/UX design practitioners when AI are being integrated into their daily life. Job skills are one of the most important demands in the working environment. It has a large effect on securing a job for any employee (Chen et al., 2024). Integrating AI systems in educational subjects, including, but not limited to, UI/UX designs, has introduced risks such as diminished creativity, over-reliance, and ethical concerns like plagiarism and data bias. Study also has shown that it often leads to reduced critical and analytical thinking skills (Zhai et al., 2024). Furthermore, operational friction is happening due to costly software and workflow overhauls (Chaudhry, 2024). Reports claim that AI agents and assistants are delivering stronger ROI in IT, sales and customer service but it still can create significant operational costs due to the needs of more orchestration, monitoring and real-time processing capabilities than generative AI-tools (Stryker & A. D., n.d.).

Despite all that, during this rapid backend challenges, a fundamental empirical gap remains, which is how the end-users actually perceive, value, and interact with AI-generated interfaces versus manually designed alternatives remains poorly understood. If this gap is being neglected, the tech industry is facing a high risk of deploying automated design pipeline that produce interfaces that users ultimately reject or distrust, making the efficiency gains of AI redundant and damaging product adoption.

3. Research Aim

To address the empirical gaps identified in the problem statement, this study aims to evaluate end-user preferences regarding the usability, aesthetic appeal, experiential quality of AI-generated versus manually designed UI/UX interfaces, and to determine how explicit user awareness of an interface's design origin influences this perception

4. Research Objectives

RO1: To compare user preferences on the visual design environment of AI-generated and manually designed UI/UX interfaces

RO2: To evaluate user preferences on the usability of AI-generated and manually designed UI/UX interfaces

RO3: To examine how the limitations of AI-assisted design tools such as content authenticity, skill dependency, and workflow disruption influence user perceptions of AI-generated UI/UX interfaces

5. Research Questions

RQ1: How do users perceive the visual design environment of AI-generated UI/UX interfaces compared to manually designed ones?

RQ2: To what extent do users prefer the usability of AI-generated UI/UX interfaces over manually designed ones

RQ3: How do AI design limitations (disinformation, skill erosion, and workflow incompatibility) affect user perceptions and preferences towards AI-generated interfaces.

6. Research Significance

From an academic perspective, this research aims to fills a critical gap in human-computer interaction in the field of feedback by end-user regarding automated design (Zhang & Gosline, 2023). Majority of nowadays study are focusing on how fast and how reliable the automated engine actually is, without actually taking notes of how the end-use actually perceive it. Hence, the data about end-user's point of view is very scarce and limited. End-user's awareness of design origin is used as the variable so that this research may find the solution of how exactly the automation bias influences user perceptions towards digital product that are designed by human versus AI .

Automation bias (AB) refers to a cognitive phenomenon where humans are over relying on automated systems and preferring automated recommendations over their own judgement while other more accurate and contradictory information is available (Romeo & Conti, 2025). AB is often considered dangerous and since it will lead to catastrophic outcomes if not adequately addressed especially when making important decisions in critical sectors. This will then dull human intuition and expertise to navigate complex or novel situations (Hoffman, 2024). This situation may occur for UI/UX designers and practitioners. When they rely too much on automation, as in artificial intelligence, their skills will erode and will often miss important requirements requested by the clients. For instance, when a designer accepts an automatically generated design entirely without double checking the requirements made by the clients, the target demographic may not be fulfilled.

Practically, this research proposes insights for digital products managers, software developer firms, and UI/UX practitioners. Since companies are forced to cover high operational costs to restructure their way of working to absorb and adopt the way of AI works, the findings of this research will explain whether the UI/UX designs that are generated using artificial intelligence actually meet the end-user needs, satisfaction, and expectations (Shukla et al., 2025). Firms that rely heavily on generative AI tools like Figma AI or Uizard tends to produce a generic, conventional layouts based on already existing patterns. While AI is rapid in prototyping, human touch and intervention is needed to ensure that complex evaluation is being made thus avoiding missing specific client needs.

7. Overview of the Proposed System

According to Figma, by using their AI website builder, there are only 4 crucial steps to creating a website, which are prompt input, layout generation, design customization, and test and adjust (Figma, n.d.). From these steps, a crucial key step is missing, which is the final check-up using an AI system itself whether the generated website follows all the prompted requirements or not. Without this step, generated websites may produce errors and even unwanted and missing features which decreases the end-user satisfaction of the generated layout and UI/UX.

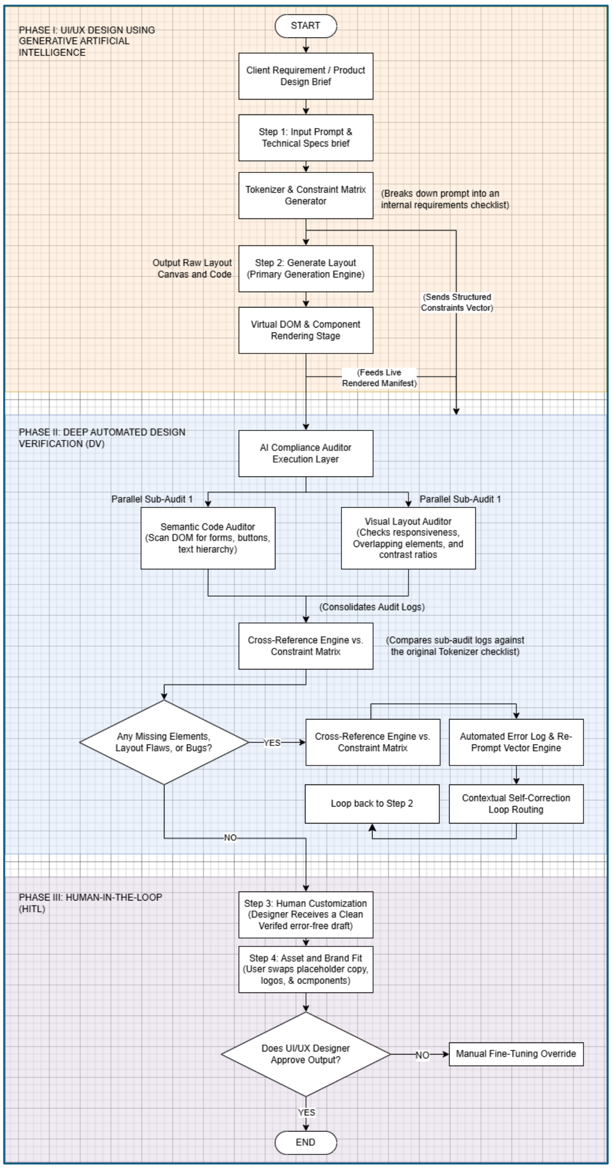

Figure 1-Proposed System Flowchart

This proposed system includes three different phases that are interrelated to each other. The phases are UI/UX Design Using Generative Artificial Intelligence (commonly offered by most of the generative AI), Deep Automated Design Verification, and lastly, Human-In-The-Loop.

Starting with the first phase, a client needs to discuss thoroughly about what is the needed requirements for the website and UI/UX design. If the clients directly feed a messy and conversational design brief to the generative AI, chances of the AI to overlook crucial details tends to become higher. This phase is designed to overcome this problem by transforming unpredictability of the human language into an absolute and unbending mathematical benchmark. When a user inputs their design prompt into the system builder, the workflow begins. Below is the example of the simple keyword prompt.

```
We need a sleek website for our law firm. It absolutely must have
a dark mode option because our clients browse late at night.
We need a 3-step consultation booking form, at least two clear
call-to-action buttons on the homepage,and it must comply with AAA
accessibility contrast standardsso it's readable for everyone
```

Figure 2 Prompt Example

After the prompt is inserted, the tokenization process take place. The prompt will be fed into the tokenizer to strip the conversational style text (e.g. “sleek website”, “late at night”) and isolate the hard engineering demands. An individual and operational feature token are now generated. Moving on to the next phase which is Semantic Parsing via the Natural Language Processing (NLP) Layer, the token is now taken and its intent are analysed by the NLP. Boolean parameters such as features that can be turned on or off or dark mode features will be identified. Apart from that, quantitative parameters and relational or boundary parameters will also be identified. Quantitative parameters refer to features that have numerical counts like 3 form steps or 2 buttons while relational or boundary parameters refer to strict mathematical thresholds like contrast ratios >= 4.5. All of these parameters will be translated into standardized UI/UX system rules. Finally, the last step will compile the constraint matrix. Constraint matrixes are constructed by the NLP extracting the rules. This matrix will serve source of truth for the entire lifecycle of the design process and it will remain permanent and unchanged.

$$\begin{bmatrix} \text{Feature ID} & \text{Feature Identifier} & \text{Type} & \text{Target Value} \\ F_{01} & \text{Dark Mode} & \text{Boolean} & 1 \\ F_{02} & \text{Contact Form} & \text{Boolean} & 1 \\ F_{03} & \text{Form Steps Count} & \text{Integer} & 3 \\ F_{04} & \text{CTA Button Count} & \text{Integer} & 2 \\ F_{05} & \text{Contrast Threshold} & \text{Float} & \geq 4.5 \end{bmatrix}$$

Figure 3 Constraint Matrix

This constraint matrix is vital for the whole process especially for the incoming phase 3 auditor. It will compare the result with this constraint matrix to make sure that all of the requirements are met. The system now achieves what the common nowadays AI cannot do which is locking the client's expectations as mathematical data. When entering Phase 2, this matrix will run quietly in the background. When entering Phase 3, this exact matrix will be used to be run the delta calculations. By implementing this concept, the machine is the one bearing the burden of tracking down whether a requirement was forgotten or vice versa, not the UI/UX designer themselves.

The Deep Automated Design Verification consists of AI compliance Auditor Execution Layer. To perform structural auditing and ensure that the generated layout follows the requirement posed by clients, this section takes the raw generated code and compiles a live virtual prototype snapshot. Two parallel sub-audits took place and carry different process. The first sub-audit performs semantic code audit (SCA). The process of understanding the code by its meaning, behaviour, and intent instead of solely on its textual structure is called semantic code analysis. Data flows, variable states, function relationships, and execution paths, are all inserted into a built model to understand how the code actually works. Furthermore, the movement of data through an application is tracked, values hold by variables at different point are noted, and functions interaction across the codebase is analysed to determine what the code actually does (apiiro, n.d.).In this proposed system, SCA will scans Document Object Model (DOM) to make sure the code aligns with the prompted requirement.

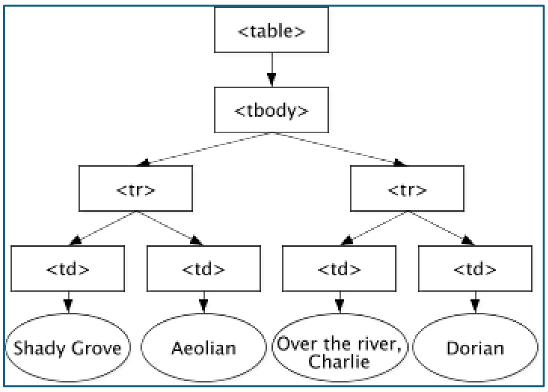

Figure 4 Document Tree

DOM refers to a tree-like structure of the webpage that will be loaded into the browser which represents the web page using a series of object (Ubah, K., 2021).

The other parallel sub-audit performs a visual layout audit. This refers to the review process of all visual design elements used by a brand or a company to make sure the design actually follow the guidelines(Kavcic, n.d.). In this proposed system, the visual layout auditor will examine and look at the rendered interface by mimicking how human eye would see it. The only difference is it will use mathematical and geometric coordinate check. A visual canvas of the UI across multiple device screen sizes (e.g. desktop, tablet, and mobile sizes) will be generated for it to be inspected by computer-vision for error detection. Computer-vision commonly referred to as artificial intelligence that specializes in measuring, recording, representing, and analysing the world from visual inputs e.g. image and video data (Kalluri et al., 2025)..

There are three core parameters that will be checked by the visual auditor. The first one is visual overlapping. CSS positioning properties sometimes are calculated incorrectly by generative AI and causes the elements to be on top of each other. The audit will run a mathematical intersection check. It will assign a geometric coordinate grid ([X, Y]) to every element's boundary. This is known as bounding box. The system will flag a collision error if the bounding box of an <h1> heading overlaps with the coordinates of other elements such as a paragraph block or the navigation bar. Next, it will check for spacing anomalies which is padding and margin consistency. Whitespace intervals will be scanned and layout asymmetry error will be flagged. Finally, the audit will also check the broken text wrapping and clipping. A website might look perfect on one display and messy on the other. Single trailing letter may drop to a new line and headlines can clip out of their containers entirely which will hide visually important text from the end-user. A responsive typography breakage will be flagged if a sentence wraps awkwardly. By adding this process, it eliminates the need of a human UI/UX designer to adjust the layout manually.

After the audit phase is done, the process moves on to the next one which is Cross-Reference Engine vs. Constraint Matrix. In the first phase, the client's messy text prompt will be transformed into a strict mathematical array of constraints which will look almost similar to below figure. This is called a constraints matrix and it represents a checklist of all of the requirements needed by the website.

Example Data Structure:

$$\begin{bmatrix} \text{Feature} & \text{Target Condition} \\ \text{Dark Mode} & 1 \\ \text{Contact Form} & 1 \\ \text{Form Steps} & 3 \\ \text{CTA Buttons} & 2 \\ \text{Contrast Ratio} & \geq 4.5 \end{bmatrix}$$

Figure 5 Constraints Matrix

Then there is this decision maker called as the Cross-Reference engine which works as data-matching processor. The engine will then execute three sequential, highly automated steps. It will first set up a whole logical comparison table. Target constraint will be compared to the audited reality. The delta will be used to show whether the status is 0 (match), -1 (Omission Error), or Flaw Detected (Compliance Error). Then, the critical logic gate will be run. If all of the delta shows a result of 0 then the engine will clear the design, bypass the correction loops, and push the layout forward into the human-in-the-loop phase. If there is any delta mismatch in occurrence, deployment pipeline will be halted to protect the flawed layout to be visible to human. Finally, instead of throwing an error code, the cross-reference engine should pinpoint the exact coordinates and tags that caused the mismatch so that the generator will understand the translated mathematical delta.

Then comes the final phase which is human-in-the-loop. In this phase, UI/UX designers will receive a clean and verified error-free draft layout generated by the generative AI. Designers will then replace all of the placeholder logos with the actual logo of client's brand or company. If the designer are fully satisfied of the output, then the process is considered complete. Otherwise, manual fine-tuning is necessary.

References

apiiro. (n.d.). *Semantic Code Analysis.* Retrieved June 5, 2026, from https://apiiro.com/glossary/semantic-code-analysis/

Chaudhry, B. M. (2024, May 11). Concerns and Challenges of AI Tools in the UI/UX Design Process: A Cross-Sectional Survey. *Conference on Human Factors in Computing Systems - Proceedings*. https://doi.org/10.1145/3613905.3650878

Chen, N., Zhao, X., & Wang, L. (2024). The Effect of Job Skill Demands Under Artificial Intelligence Embeddedness on Employees' Job Performance: A Moderated Double-Edged Sword Model. *Behavioral Sciences*, *14*(10). https://doi.org/10.3390/bs14100974

Stryker, C., & A. D. (n.d.). *The biggest AI adoption challenges for 2026*. Retrieved June 5, 2026, from https://www.ibm.com/think/insights/ai-adoption-challenges

Figma. (n.d.). *How to use AI to create a website. Retrieved from Figma*. Retrieved June 5, 2026, from https://www.figma.com/resource-library/how-to-use-ai-to-create-a-website/

Hoffman, B. (2024, March 10). *Automation Bias: What It Is And How To Overcome It*. Forbes. https://www.forbes.com/sites/brycehoffman/2024/03/10/automation-bias-what-it-is-and-how-to-overcome-it/

Kalluri, P. R., Agnew, W., Cheng, M., Owens, K., Soldaini, L., & Birhane, A. (2025). Computer-vision research powers surveillance technology. *Nature*, *643*(8070). https://doi.org/10.1038/s41586-025-08972-6

Kavcic, R. (n.d.). *What is a design audit and how do you conduct one?* Design_Strategy. Retrieved June 5, 2026, from https://designstrategy.guide/what-is-a-design-audit-and-how-do-you-conduct-one/#design-audit

Kim, T. S., John Ignacio, M., Yu, S., Jin, H., & Kim, Y. G. (2024). UI/UX for Generative AI: Taxonomy, Trend, and Challenge. *IEEE Access*, *12*, 179891–179911. https://doi.org/10.1109/ACCESS.2024.3502628

Ubah, K. (2021, May 17). *The DOM Explained for Beginners – How the Document Object Model Works*. FreeCodeCamp. https://www.freecodecamp.org/news/dom-explained-everything-you-need-to-know-about-the-document-object-model/

Marushchak, A., Petrov, S., & Khoperiya, A. (2024). Countering AI-powered disinformation through national regulation: learning from the case of Ukraine. In *Frontiers in Artificial Intelligence* (Vol. 7). https://doi.org/10.3389/frai.2024.1474034

Romeo, G., & Conti, D. (2025). Exploring automation bias in human–AI collaboration: a review and implications for explainable AI. *AI and Society*. https://doi.org/10.1007/s00146-025-02422-7

Shukla, P., Bui, P., Levy, S. S., Kowalski, M., Baigelenov, A., & Parsons, P. (2025). De-skilling, Cognitive Offloading, and Misplaced Responsibilities: Potential Ironies of AI-Assisted Design. *Conference on Human Factors in Computing Systems - Proceedings* . https://doi.org/10.1145/3706599.3719931

Zhai, C., Wibowo, S., & Li, L. D. (2024). The effects of over-reliance on AI dialogue systems on students' cognitive abilities: a systematic review. *Smart Learning Environments*, *11*(1). https://doi.org/10.1186/s40561-024-00316-7

Zhang, Y., & Gosline, R. (2023). Human favoritism, not AI aversion: People's perceptions (and bias) toward generative AI, human experts, and human–GAI collaboration in persuasive content generation. *Judgment and Decision Making*, *18*. https://doi.org/10.1017/JDM.2023.37